Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – VIII

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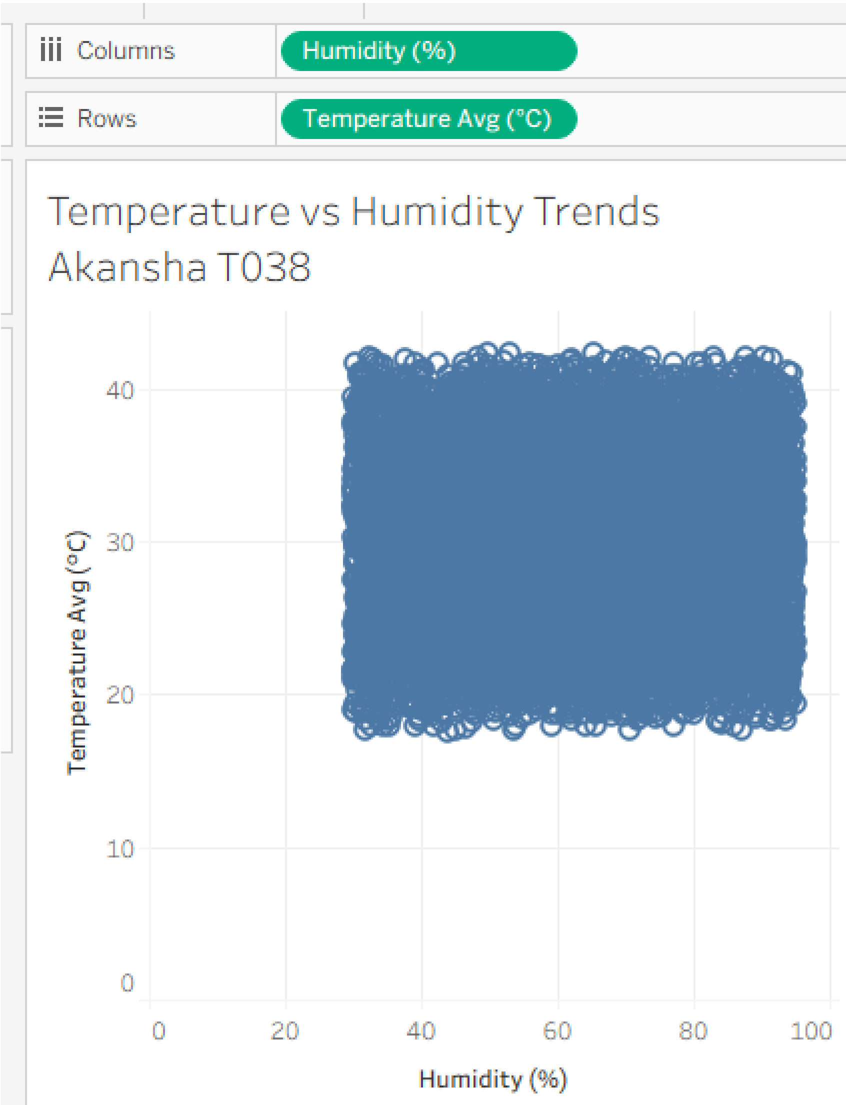
Class: TYIT	Batch: 1
Date of Assignment: 27-08-2026	Date/Time of Submission: 27-08-2026

Part B – Program

Performing Advanced Analytics using Trends, Clustering, and Forecasting

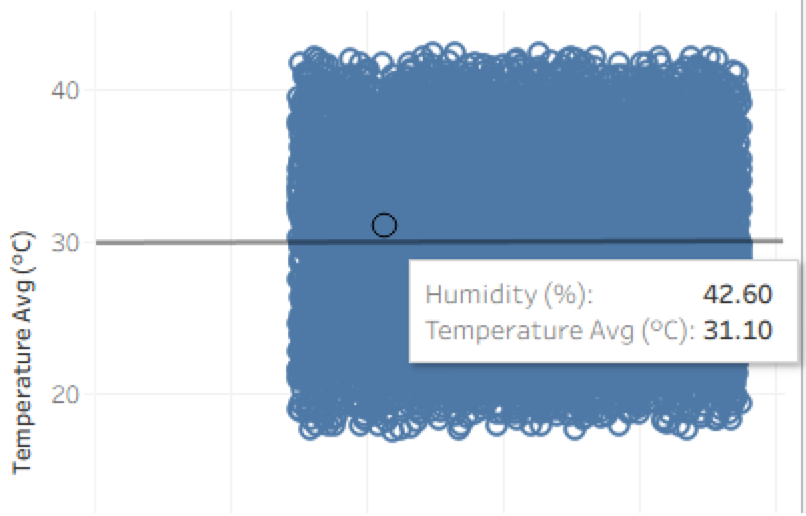
a) Add and interpret trend lines (linear, exponential)

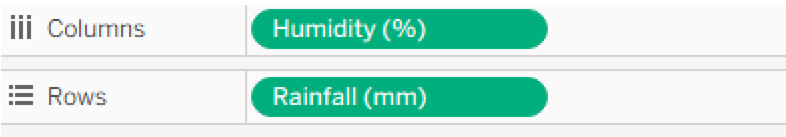
analysis-> uncheck aggregate measure



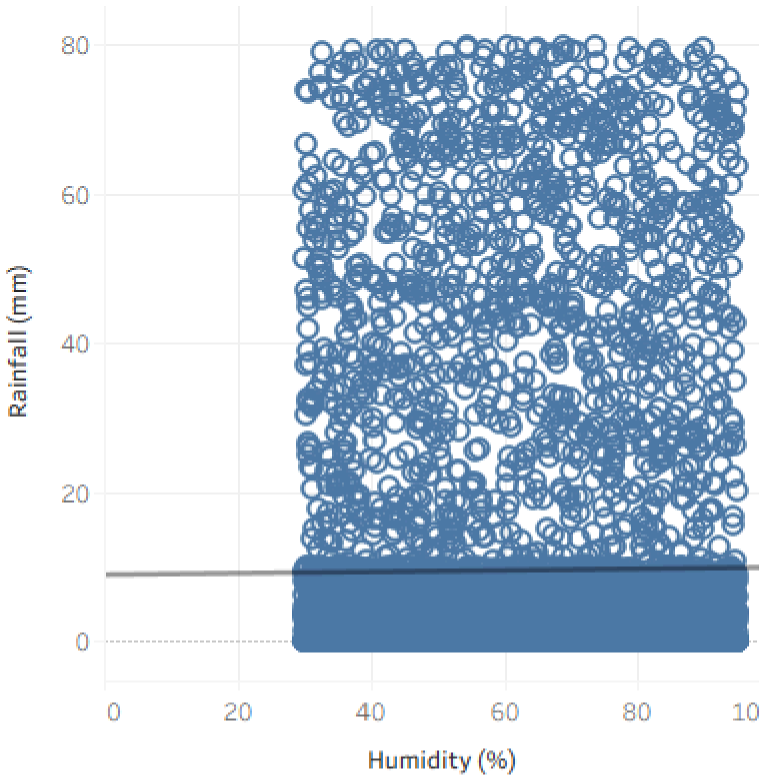
drag trend line on the canvas-> linear

Temperature vs Humidity Trends
Akansha T038

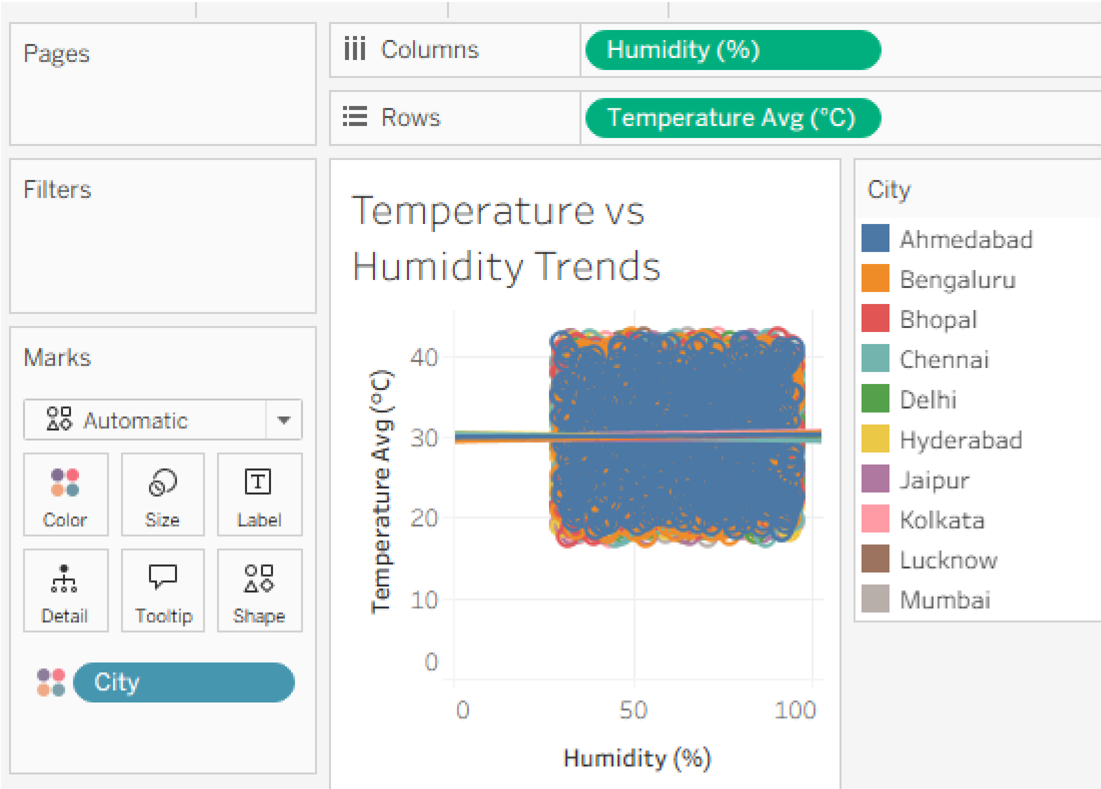


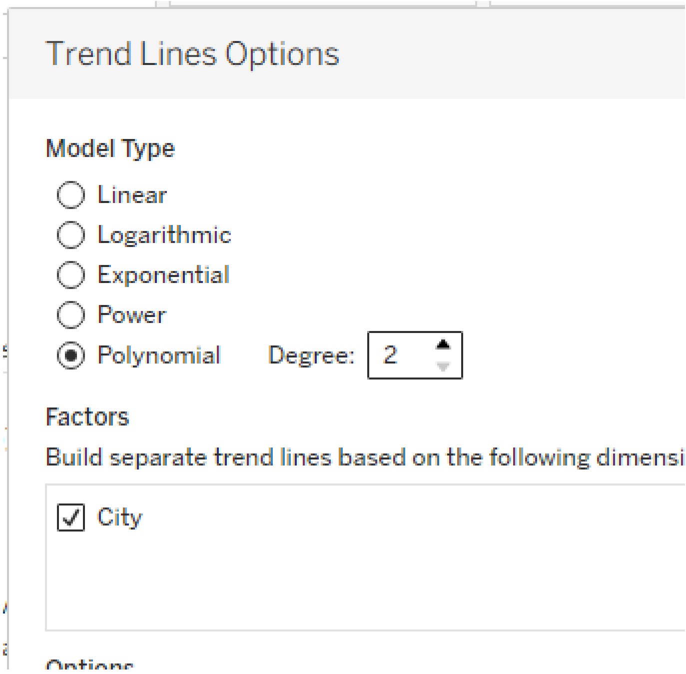


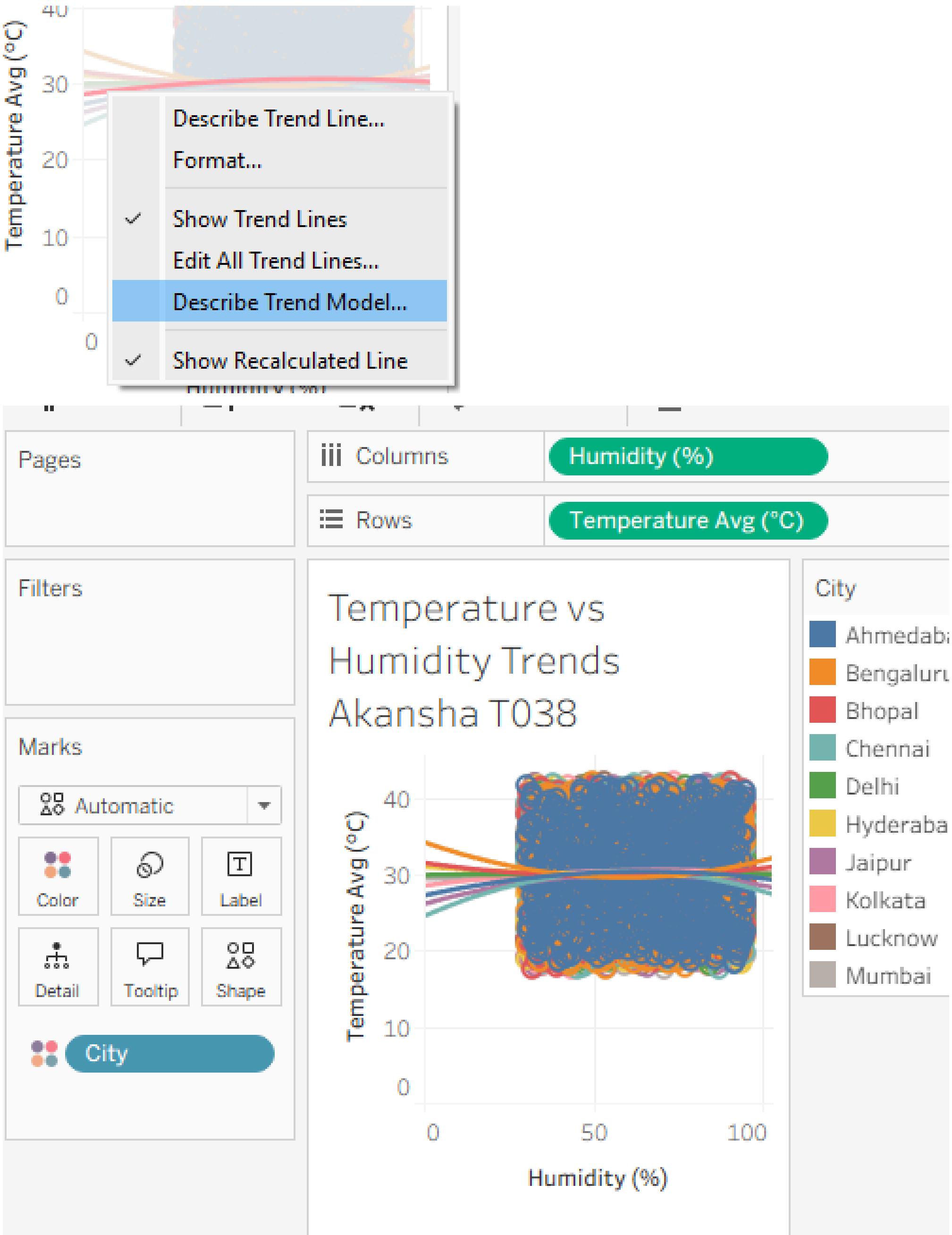
Humidity vs Rainfull
Akansha T038

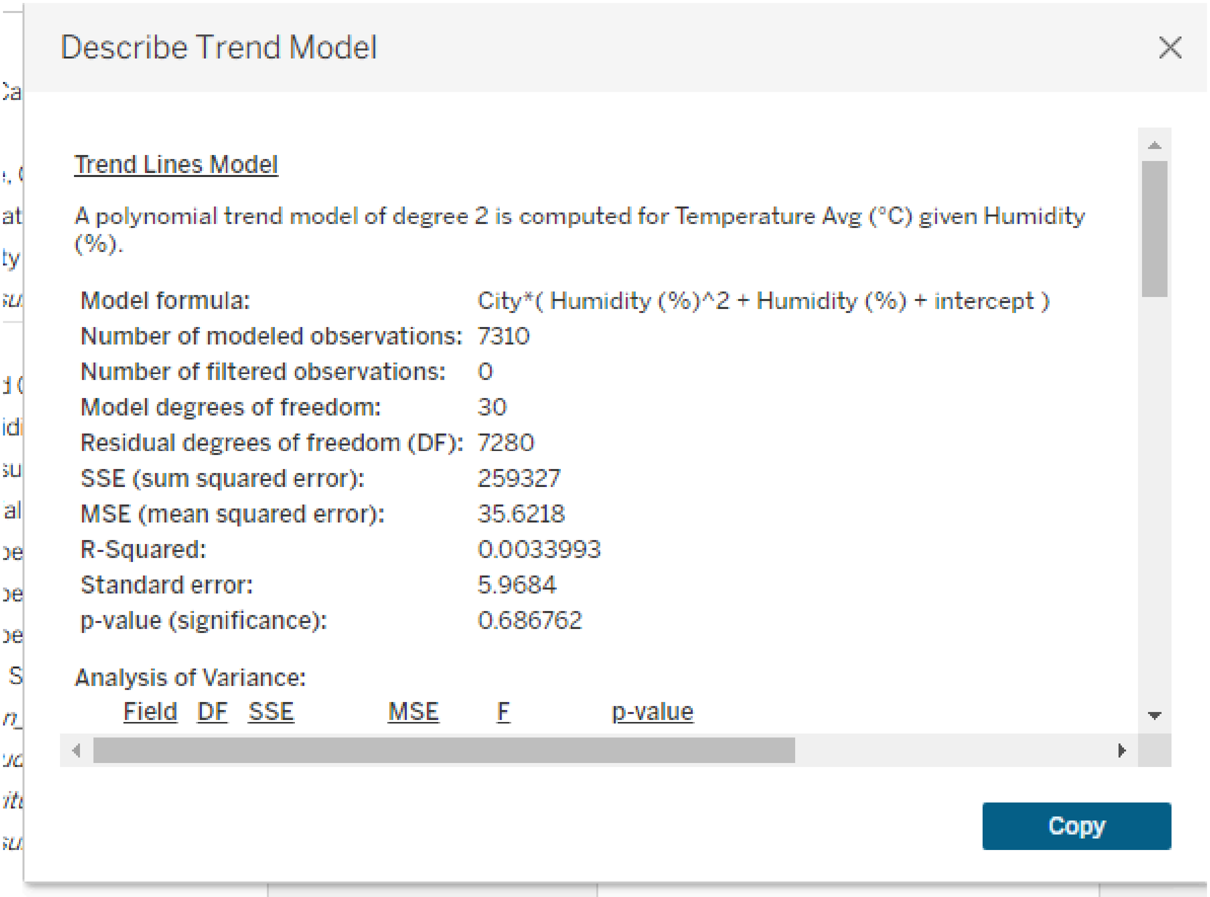


b) Customize and analyze trend models

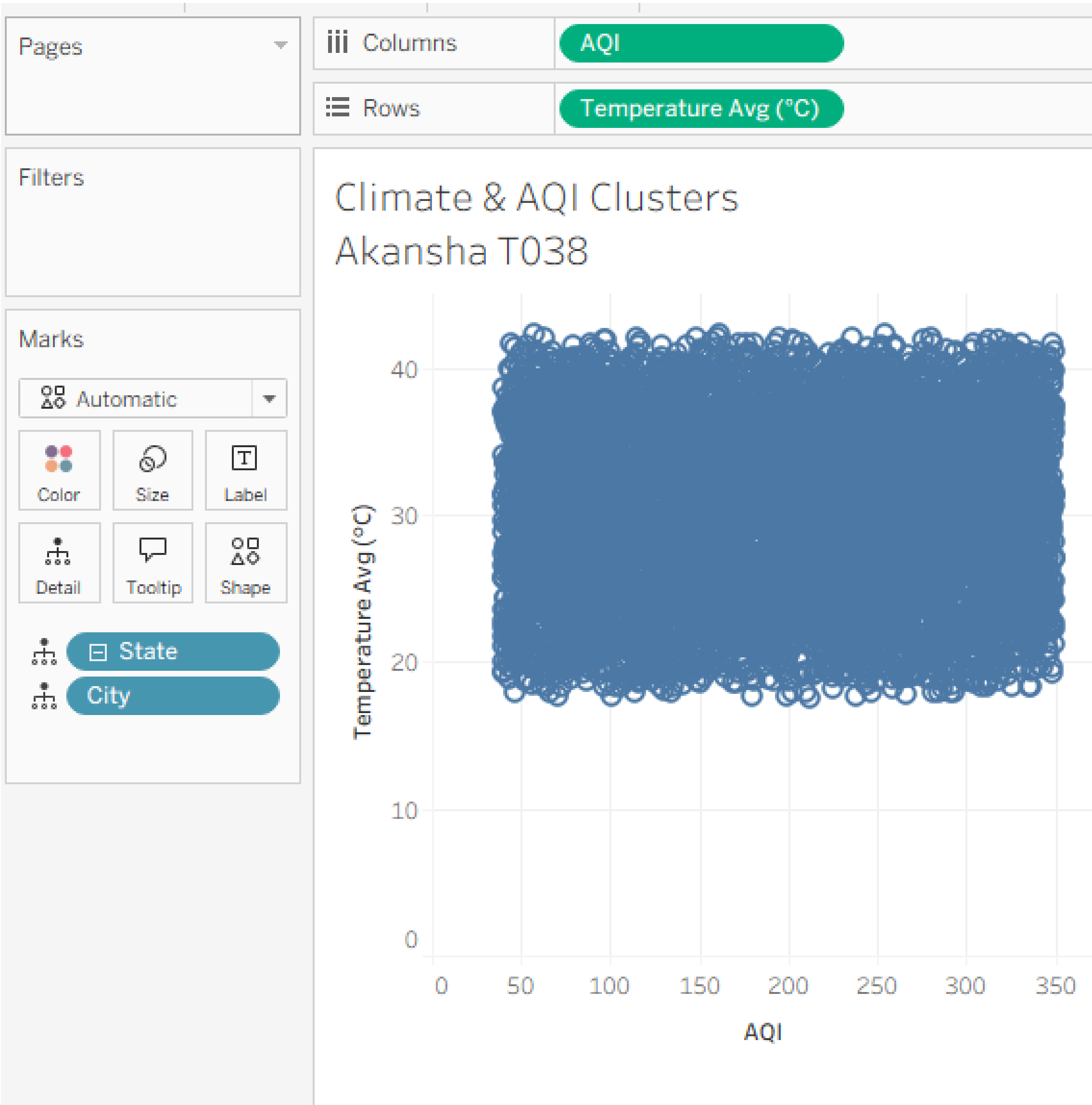


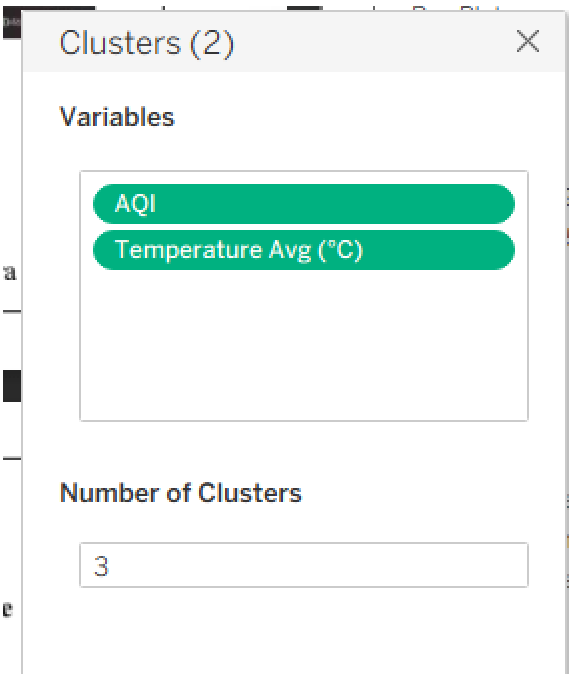


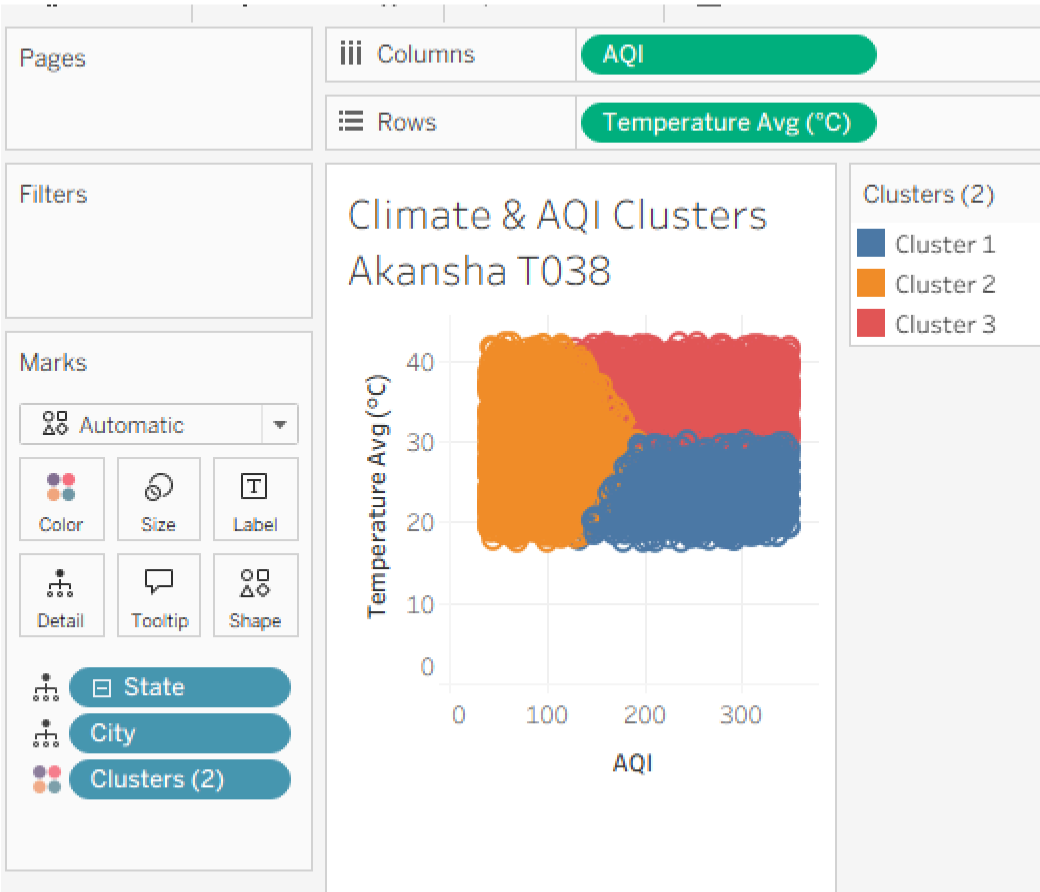

Describe Trend Line...
Format...
Show Trend Lines
Edit All Trend Lines...
Describe Trend Model...
Show Recalculated Line
Pages
Columns
Humidity (%)
Rows
Temperature Avg (°C)
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Shape
City
Temperature vs
Humidity Trends
Akansha T038
Temperature Avg (°C)
Humidity (%)
City
Ahmedab
Bengaluru
Bhopal
Chennai
Delhi
Hyderaba
Jaipur
Kolkata
Lucknow
Mumbai

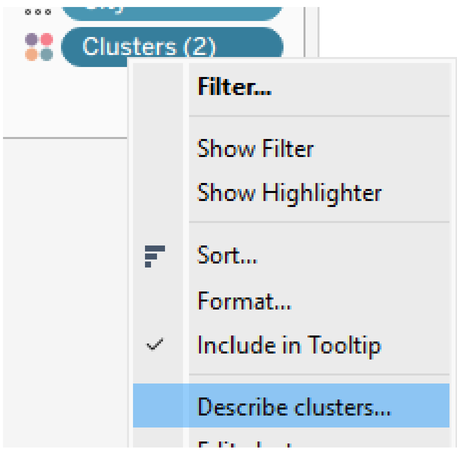


c) Perform clustering on datasets


Pages
Columns
AQI
Rows
Temperature Avg (°C)
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Shape
State
City
Climate & AQI Clusters
Akansha T038
Temperature Avg (°C)
40
30
20
10
0
0
50
100
150
200
250
300
350
AQI


Clusters (2)
Variables
AQI
Temperature Avg (°C)
Number of Clusters
3


Pages
Columns
AQI
Rows
Temperature Avg (°C)
Filters
Climate & AQI Clusters
Akansha T038
Clusters (2)
Cluster 1
Cluster 2
Cluster 3
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Shape
State
City
Clusters (2)
Temperature Avg (°C)
40
30
20
10
0
0
100
200
300
AQI


Clusters (2)
Filter...
Show Filter
Show Highlighter
Sort...
Format...
Include in Tooltip
Describe clusters...

Describe Clusters

Summary | Models

Inputs for Clustering

Variables: Sum of AQI
Sum of Temperature Avg (°C)
Level of Detail: City, State
Scaling: Normalized

Summary Diagnostics

Number of Clusters: 3
Number of Points: 7310
Between-group Sum of Squares: 635.18
Within-group Sum of Squares: 400.17
Total Sum of Squares: 1035.3

		Centers	
Clusters	Number of Items	Sum of AQI	Sum of Temperature Avg (°C)
Cluster 1	2159	258.69	24.808
Cluster 2	3053	104.75	29.835
Cluster 3	2098	256.47	35.514
Not Clustered	0		

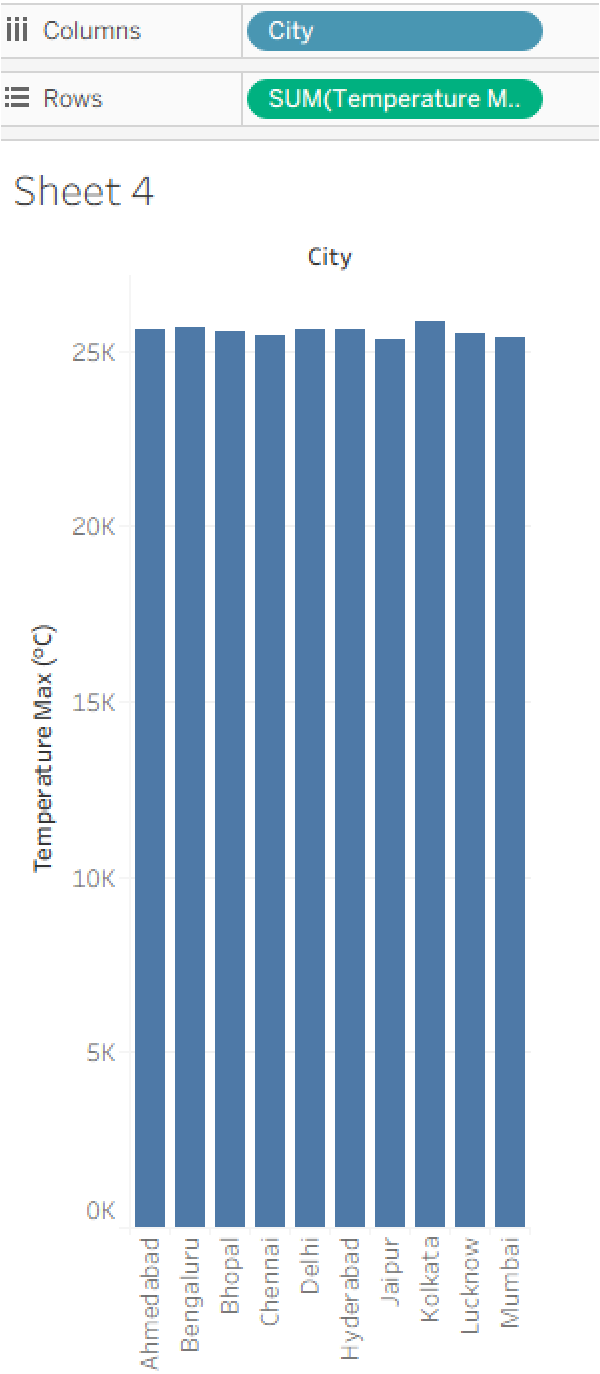
Describe Clusters

Summary | Models

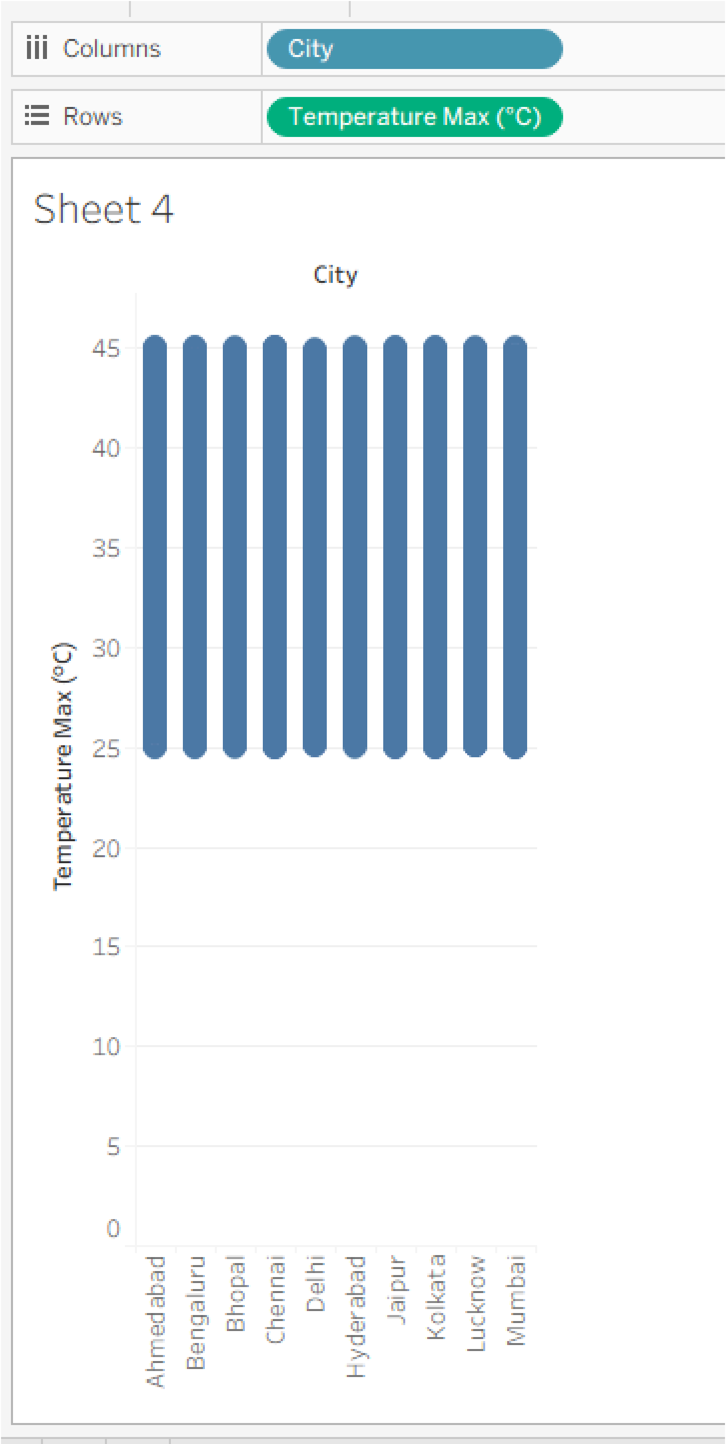
Analysis of Variance:

			Model		Error	
Variable	F-statistic	p-value	Sum of Squares	DF	Sum of Squares	DF
Sum of AQI	2611.0	0.0	435.1	2	608.8	7307
Sum of Temperature Avg (°C)	1714.0	0.0	200.1	2	426.5	7307

d) Analyze distributions using built-in tools

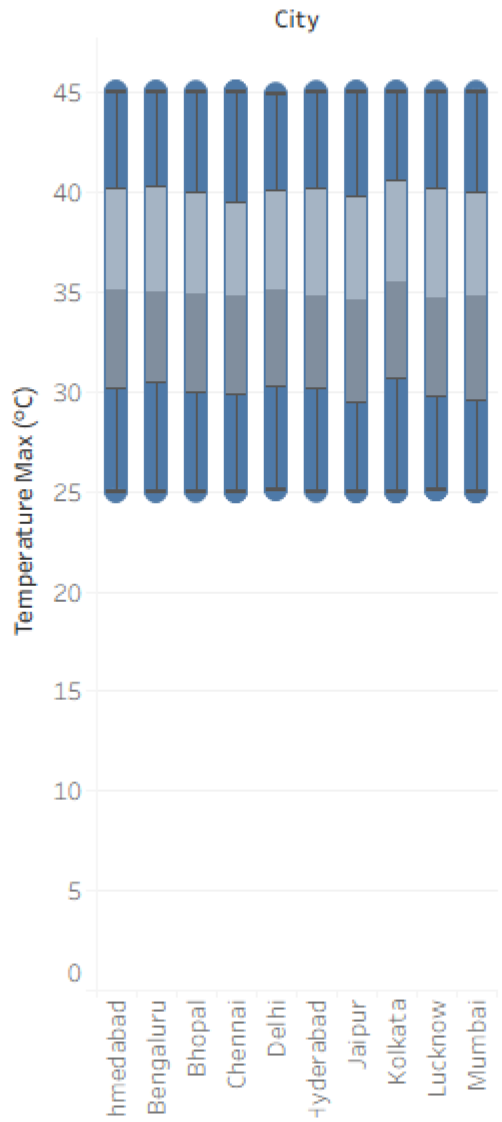


analysis-> aggregate measure

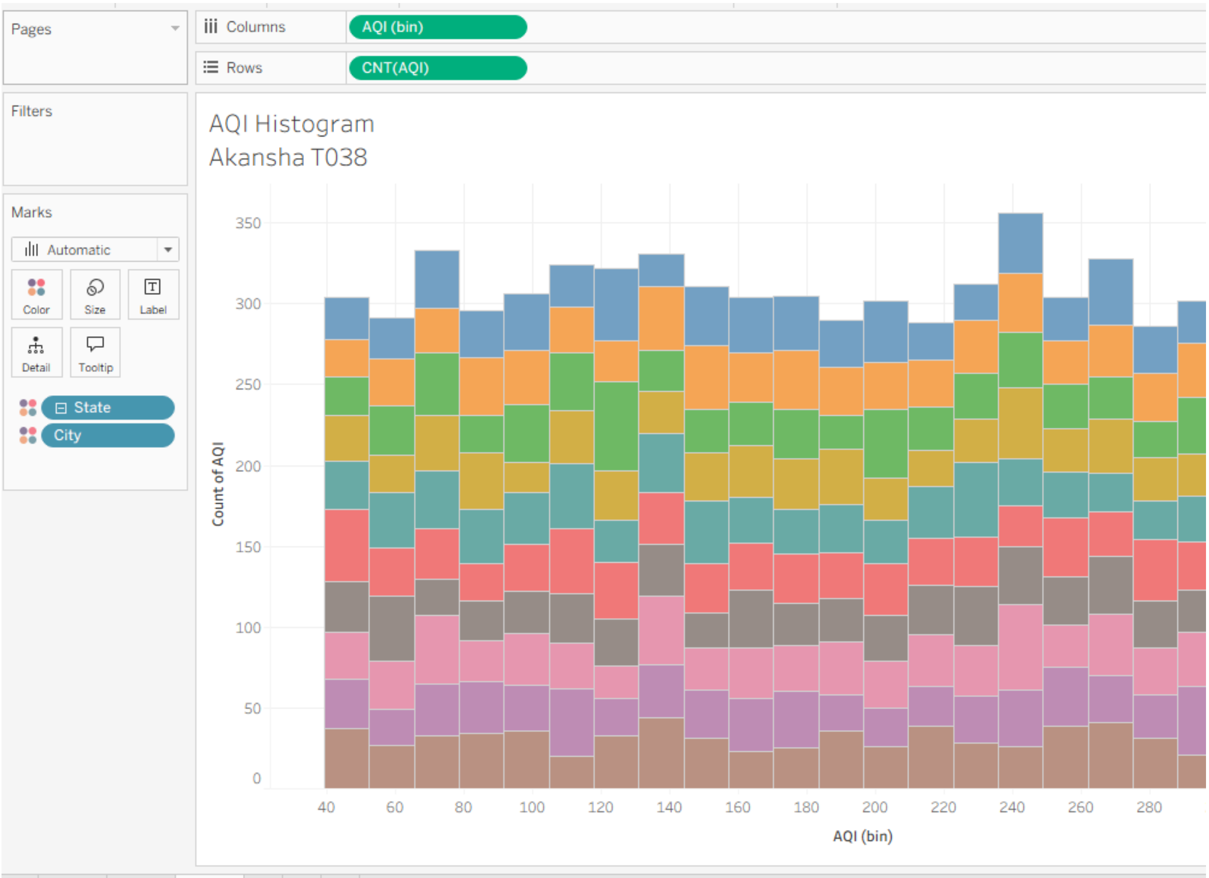


drag box plot on canvas

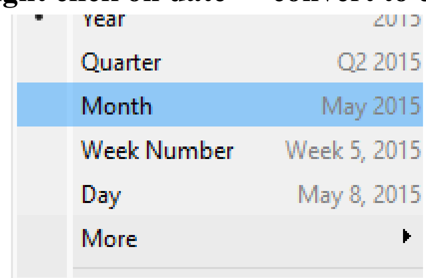
Sheet 4

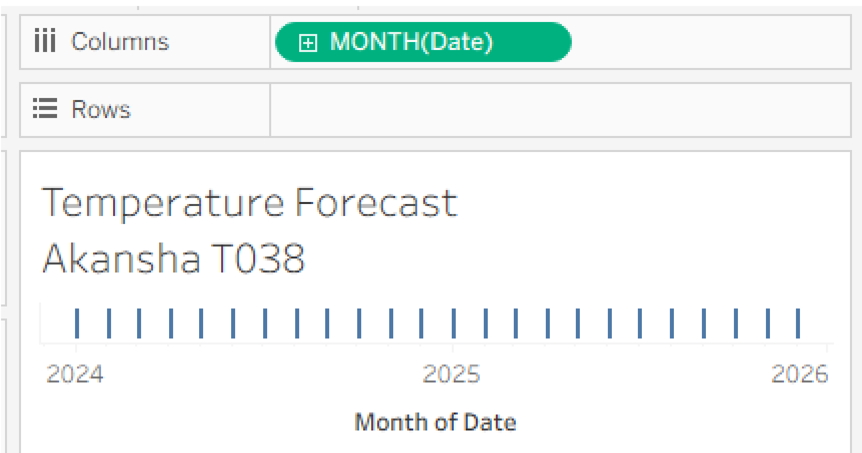


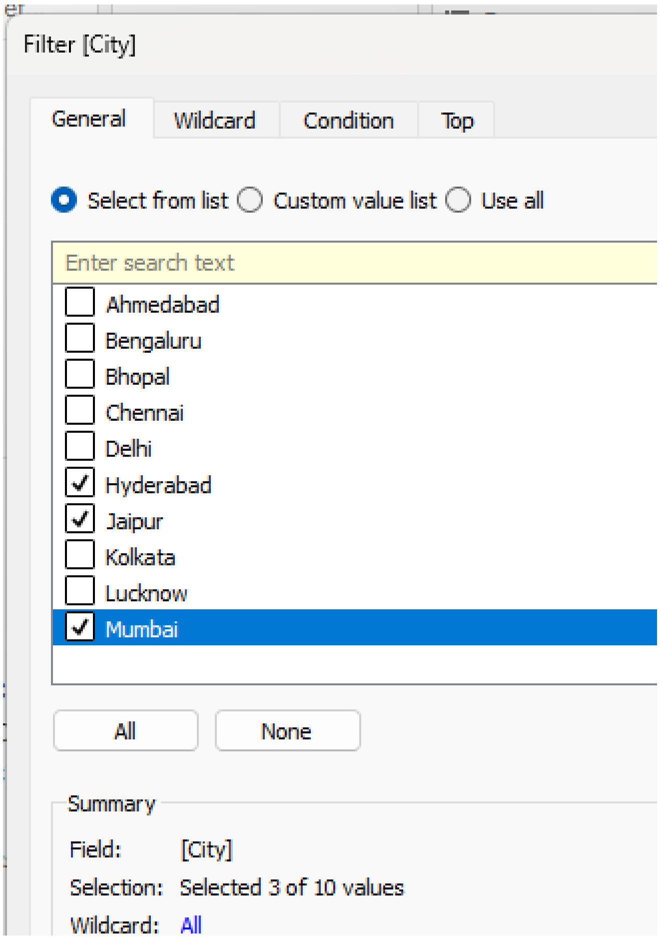
City
Temperature Max (ºC)
45
40
35
30
25
20
15
10
5
0
hmedabad
Bengaluru
Bhopal
Chennai
Delhi
Hyderabad
Jaipur
Kolkata
Lucknow
Mumbai

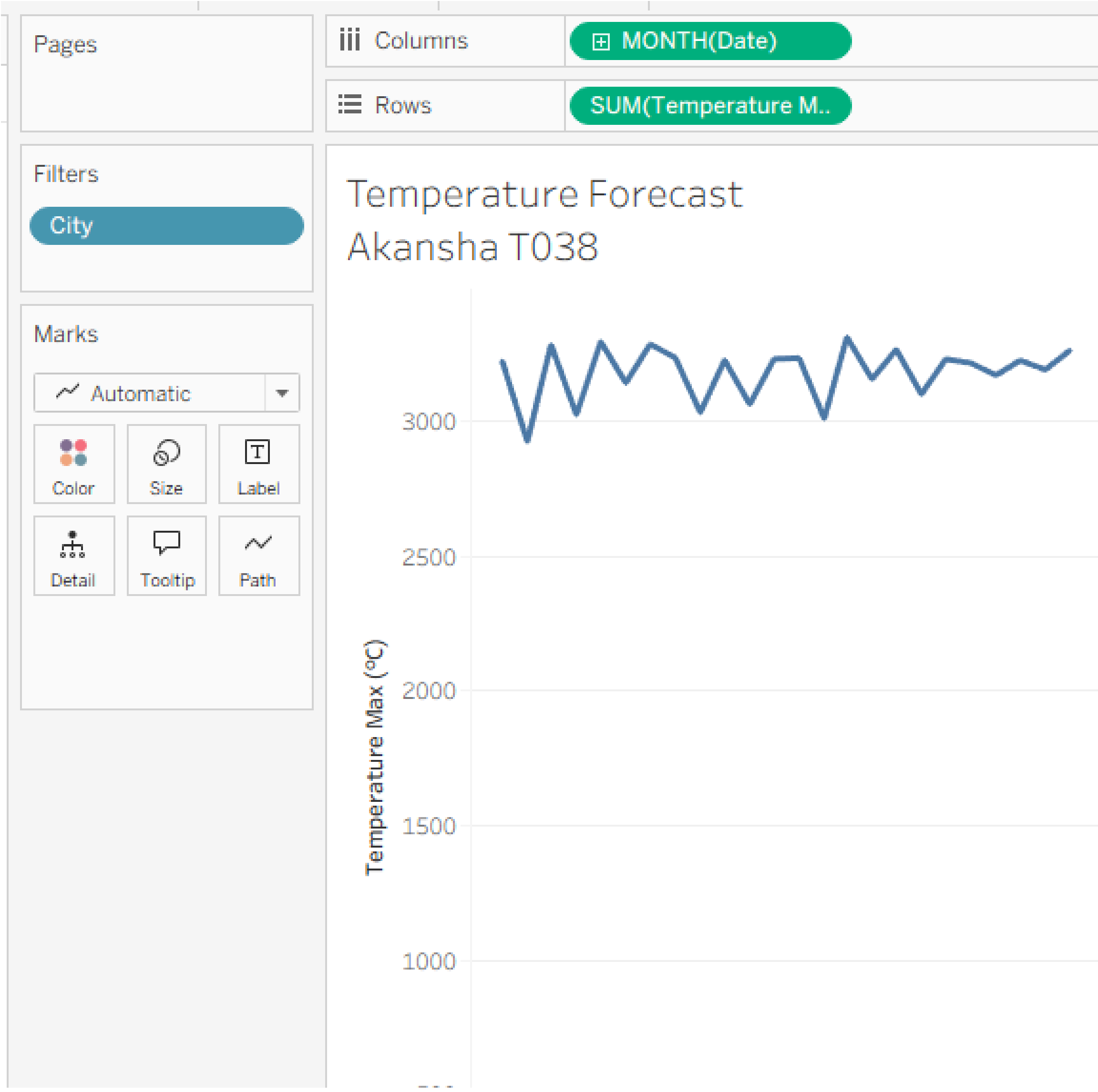


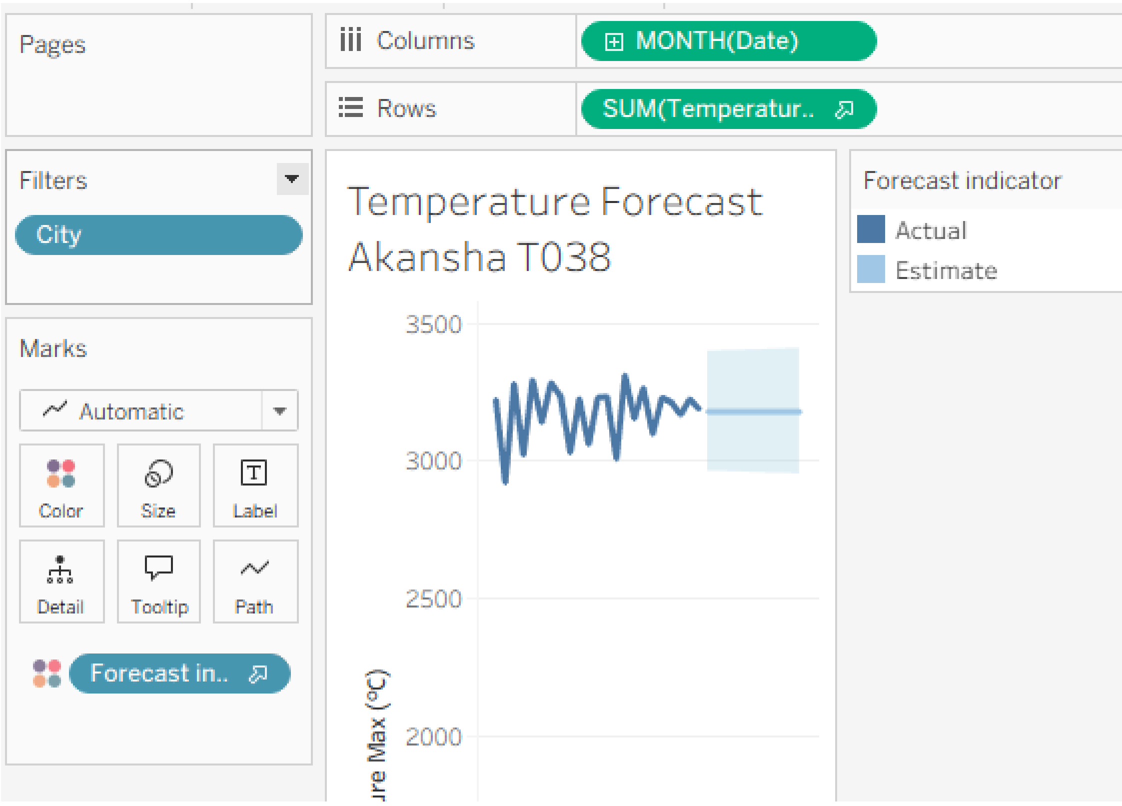
e) Apply forecasting techniques
right click on date -> convert to contiuous

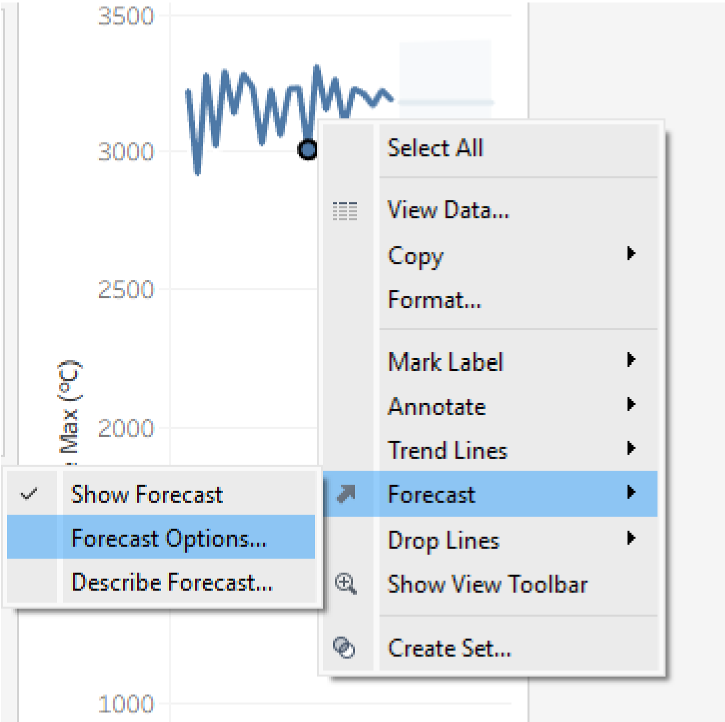


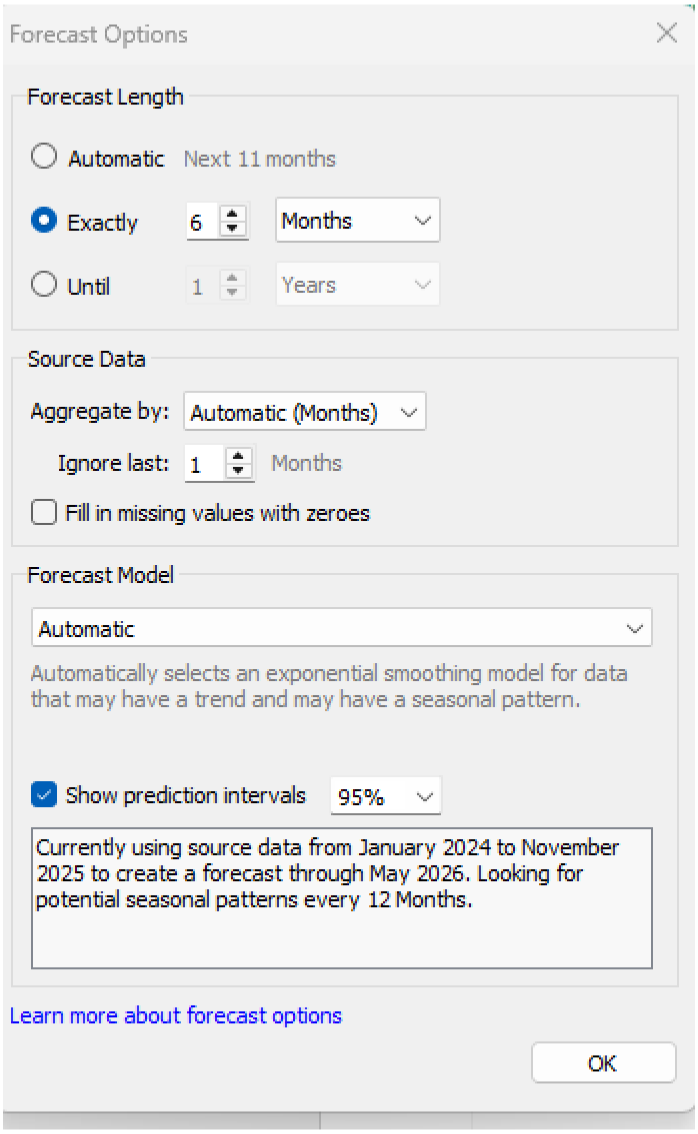


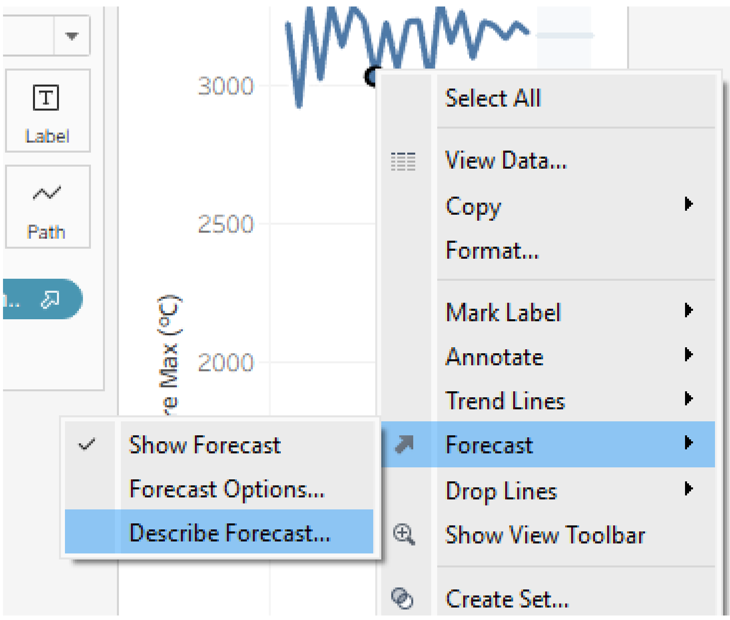

Filter [City]
General
Wildcard
Condition
Top
Select from list
Custom value list
Use all
Enter search text
Ahmedabad
Bengaluru
Bhopal
Chennai
Delhi
Hyderabad
Jaipur
Kolkata
Lucknow
Mumbai
All
None
Summary
Field: [City]
Selection: Selected 3 of 10 values
Wildcard: All

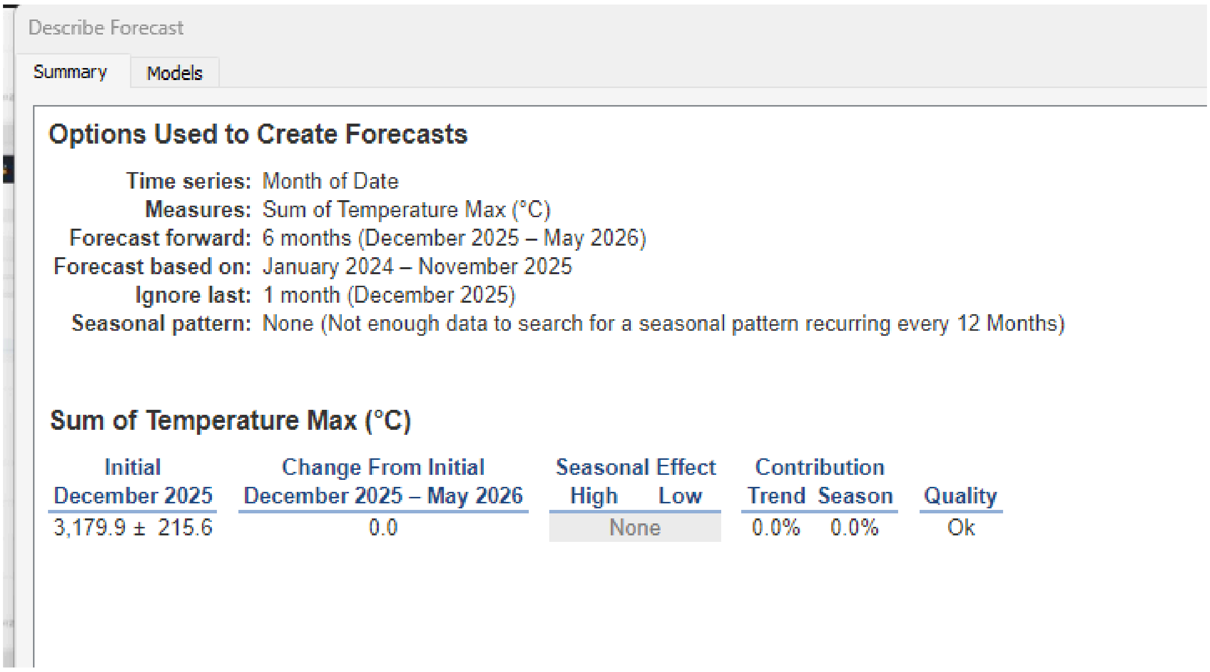

Pages
Columns
MONTH(Date)
Rows
SUM(Temperature M..
Filters
City
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Path
Temperature Forecast
Akansha T038
Temperature Max (ºC)
3000
2500
2000
1500
1000


Pages
Columns
MONTH(Date)
Rows
SUM(Temperatur..
Filters
City
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Path
Forecast in..
Temperature Forecast
Akansha T038
3500
3000
2500
2000
Forecast indicator
Actual
Estimate

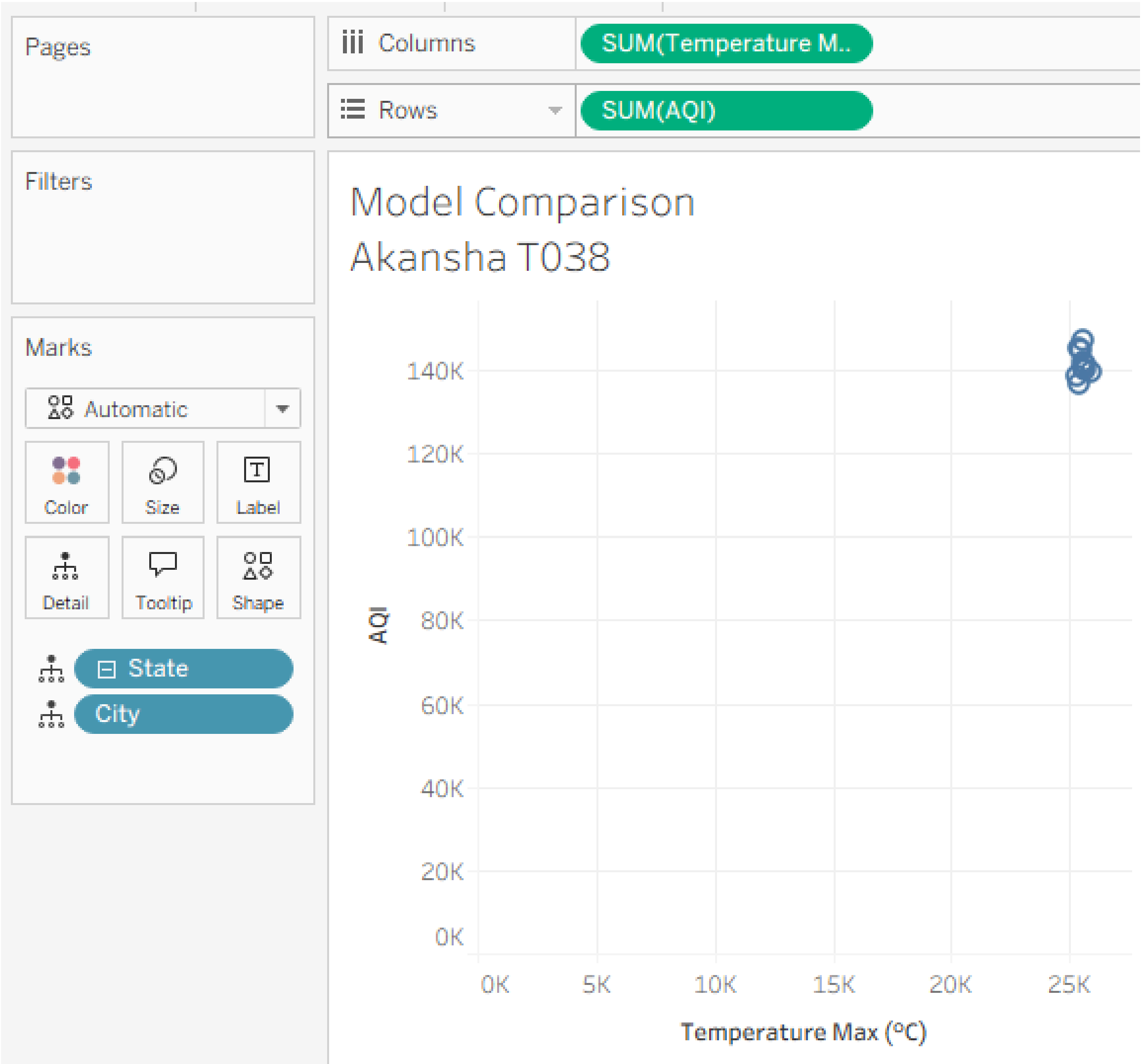

3500
3000
2500
2000
1000
Select All
View Data...
Copy
Format...
Mark Label
Annotate
Trend Lines
Forecast
Drop Lines
Show View Toolbar
Create Set...
Show Forecast
Forecast Options...
Describe Forecast...

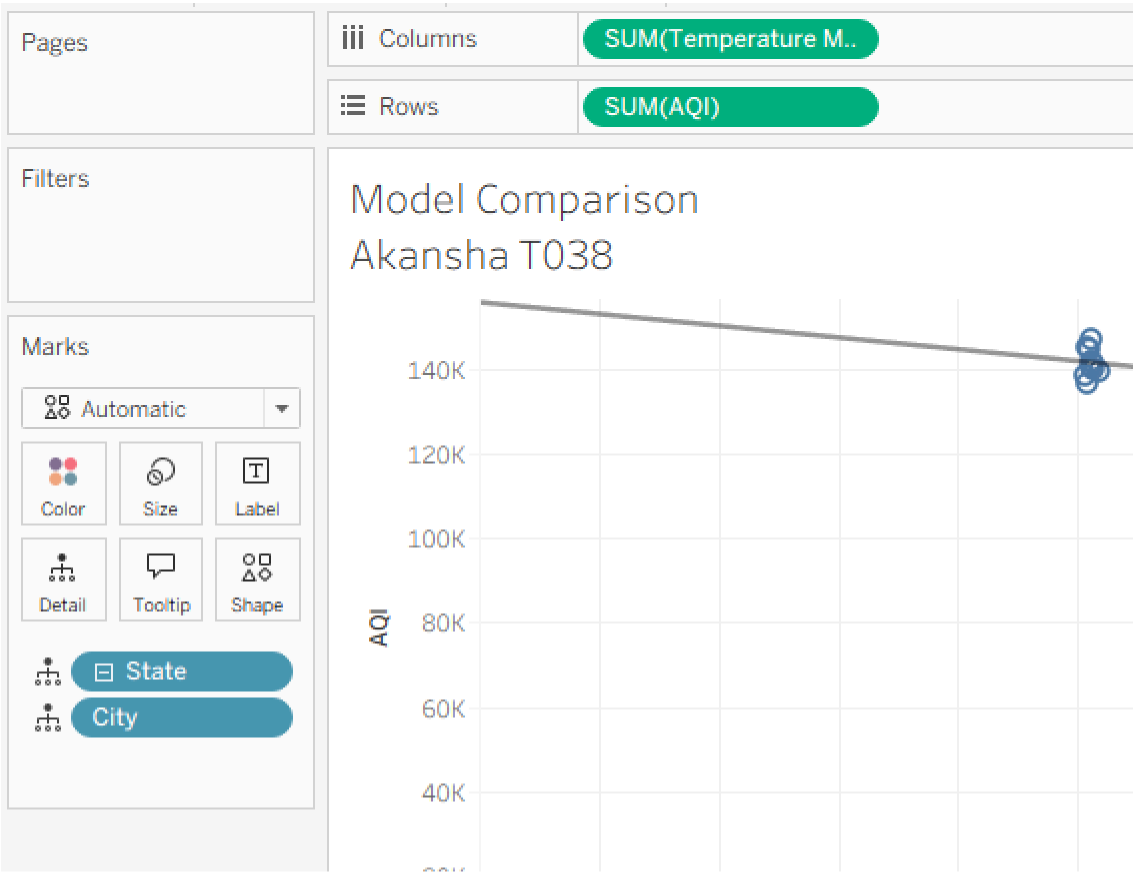

Forecast Options
Forecast Length
Automatic
Next 11 months
Exactly
6
Months
Until
1
Years
Source Data
Aggregate by:
Automatic (Months)
Ignore last:
1
Months
Fill in missing values with zeroes
Forecast Model
Automatic
Automatically selects an exponential smoothing model for data that may have a trend and may have a seasonal pattern.
Show prediction intervals
95%
Currently using source data from January 2024 to November 2025 to create a forecast through May 2026. Looking for potential seasonal patterns every 12 Months.
Learn more about forecast options
OK

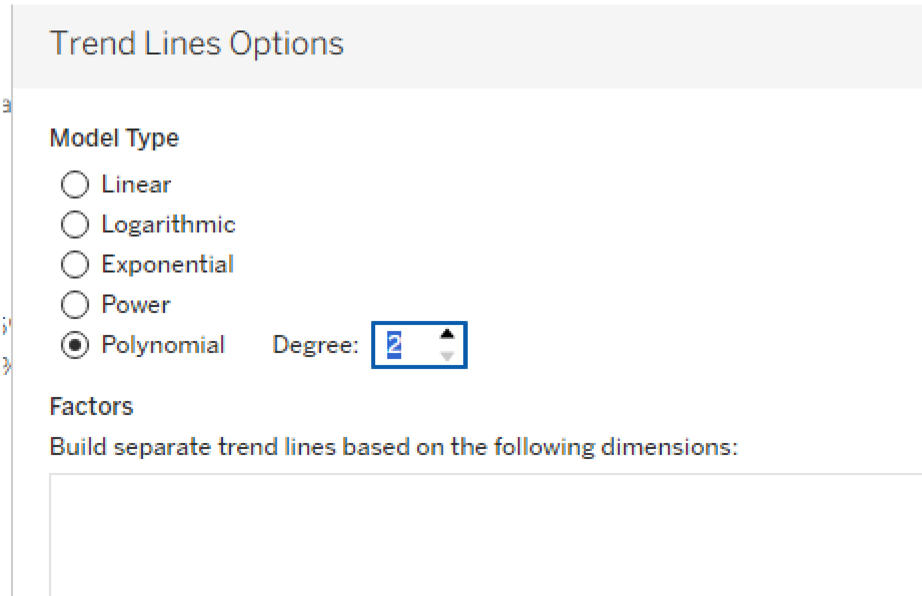

Label
Path
3000
2500
2000
Select All
View Data...
Copy
Format...
Mark Label
Annotate
Trend Lines
Forecast
Drop Lines
Show View Toolbar
Create Set...
Show Forecast
Forecast Options...
Describe Forecast...



f) Compare different statistical models







g) Export and interpret statistical results